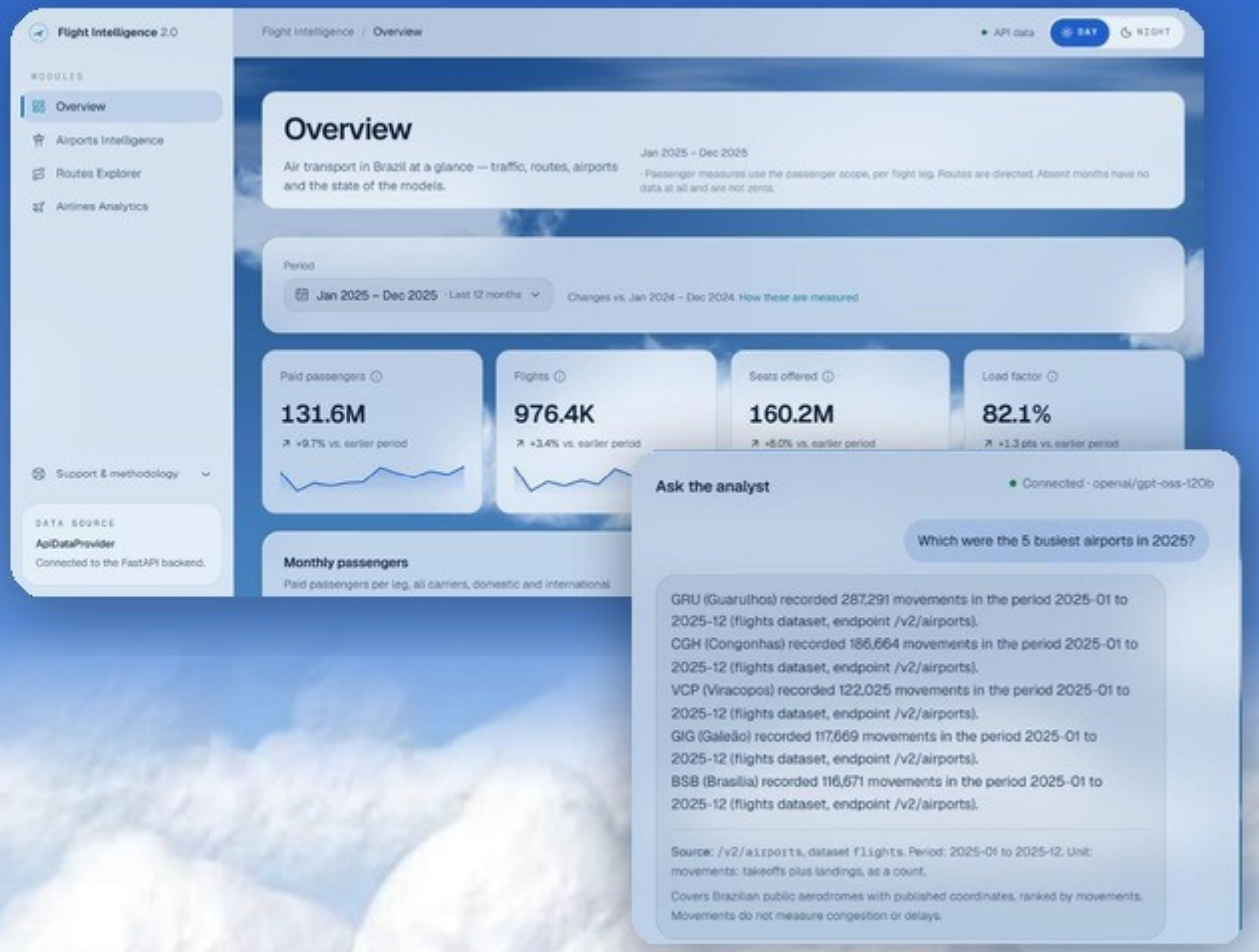
ANAC FLIGHT INTELLIGENCE

Brazilian aviation *in data,* verified.

22.1M flight legs, 2000–2025
Models tested against baselines · AI analyst

Python · Polars · DuckDB | LightGBM | FastAPI · Next.js | Groq



anac-flight-intelligence-api-main.vercel.app

Caio Bandeira · Data Science and Machine Learning (UniCEUB)

anac-flight-intelligence-api-main.vercel.app · github.com/CaioBandeira7/anac-flight-intelligence-api-main

What it is and why it matters

20 GB

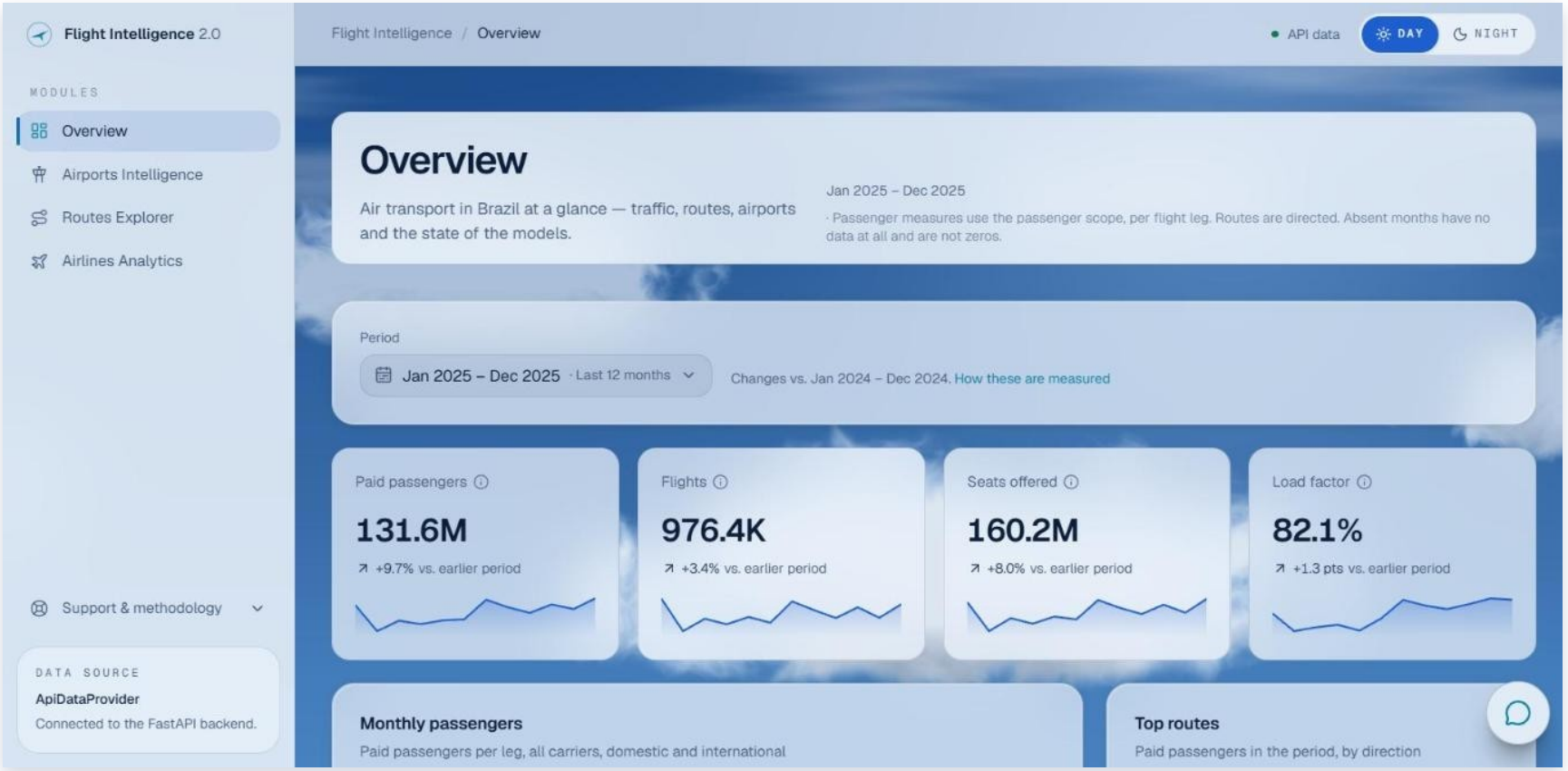
ANAC open data, every Brazilian flight leg since 2000

22.1M

rows ingested and audited, none rejected

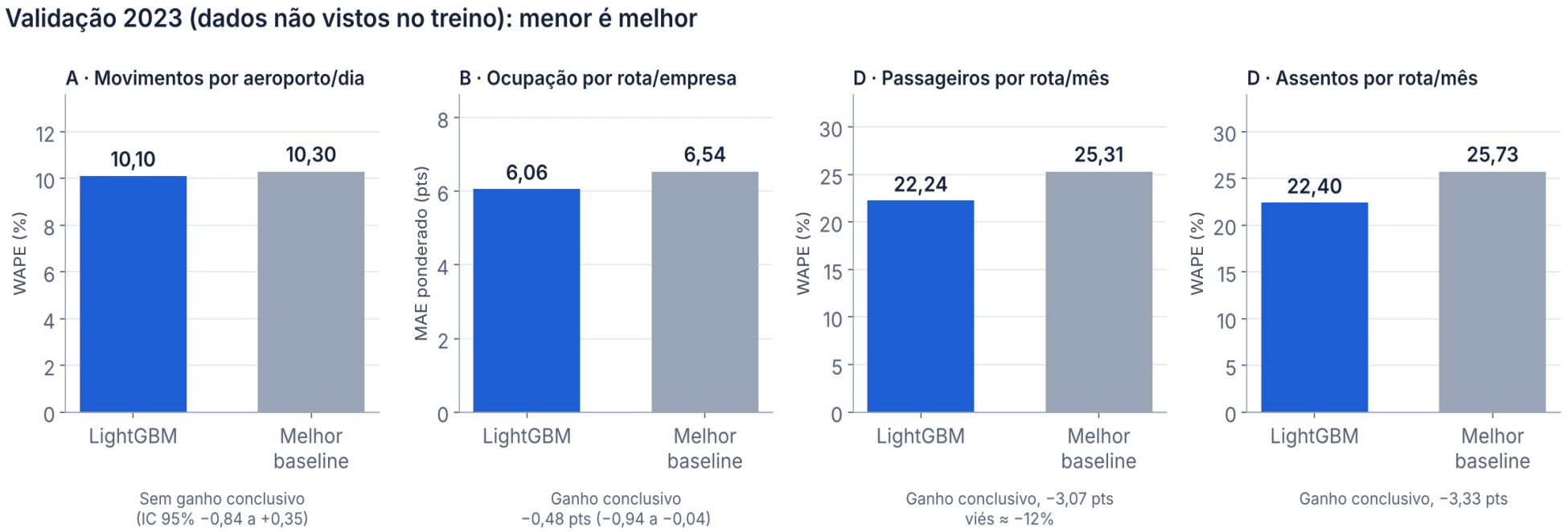
308

months with data, 2000-01 to 2025-12 (4 absent)



Pipeline: audited Parquet ingestion → point-in-time features → LightGBM vs baselines → star schema → read-only FastAPI (Render, Docker) → Next.js dashboard (Vercel) → AI analyst (Groq).

Machine learning, honestly evaluated

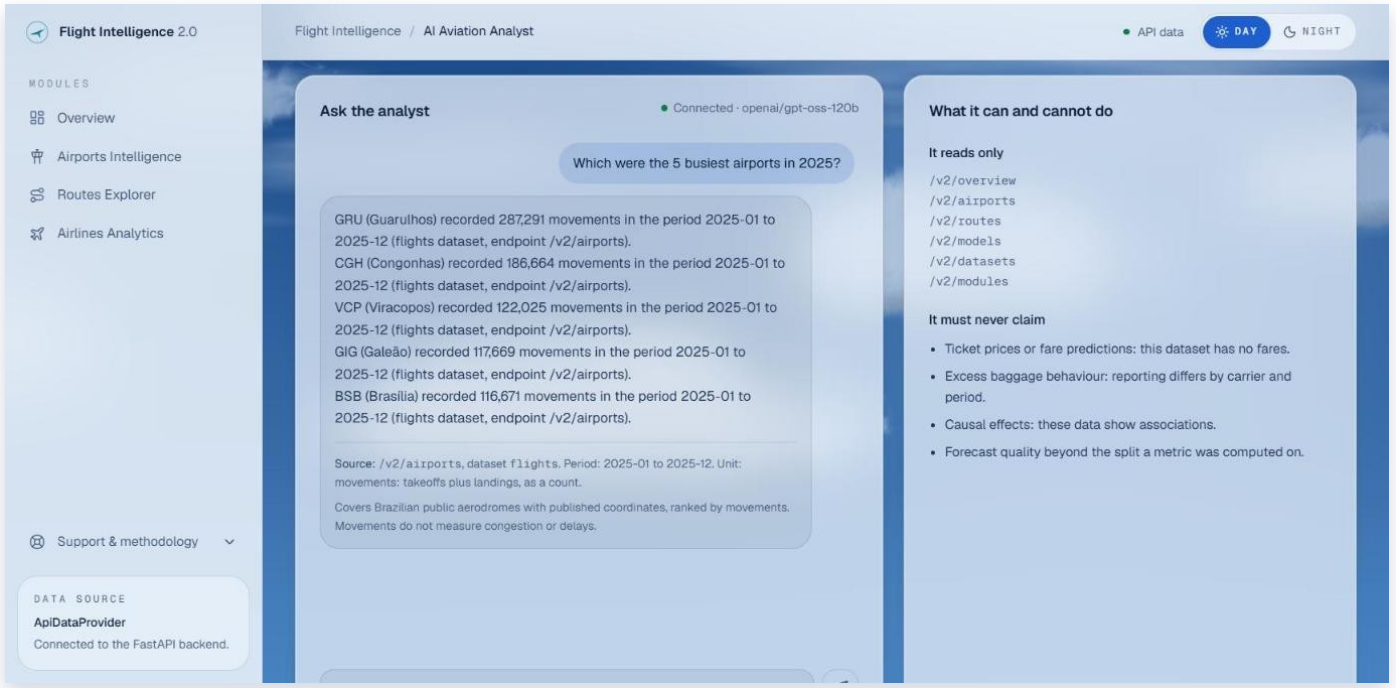


Fonte: research/ml-showcase/REPORT.md e results/analysis.json. Resultados de validação, não previsões; o teste 2024–2025 não foi executado. No módulo A o baseline é o método adotado.

- 2023 validation on unseen data; adopted only if it beats the best baseline with a 95% interval excluding zero
- Route demand, 1 month ahead: WAPE 18.7% vs 24.1%
- Airport traffic: no conclusive gain, so the simple rule is kept
- Limits: about −12% bias on route demand, worse than baseline at 6 months, 2024–2025 test not run

Labels in the chart are in Portuguese. Source: research/ml-showcase/REPORT.md. Validation results, not forecasts.

AI analyst and live demo



- Answers only through 5 fixed, read-only tools; every number is checked against the tool result before it is shown
- Sources, period and unit attached by code, not by the model
- Groq key server-side only; 10 questions per visitor per hour, 200 per day overall
- Live: anac-flight-intelligence-api-main.vercel.app

Real answer on 2026-11-08, matching /v2/airports. The numeric check is heuristic.